

Module 3: Mount NFS Volume

Exercise 1: Configuring Networking and NFS Service

In this exercise, you create a data LIF and default route, and ping the data LIF from the Linux client. Later, you verify the status of the remote procedure call (RPC) service.

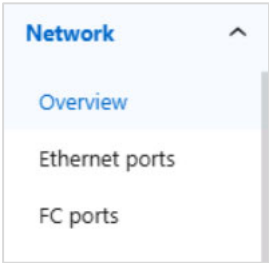
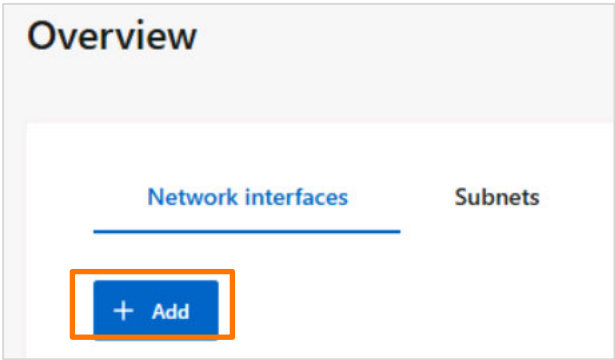
Objectives

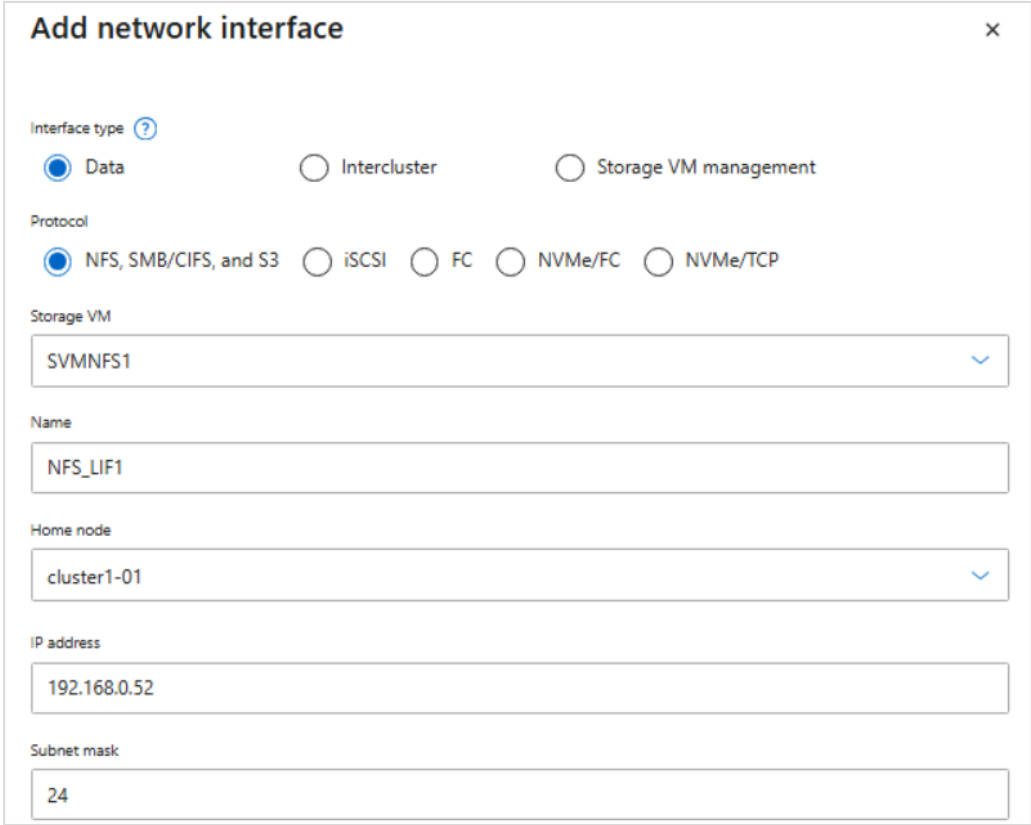
This exercise focuses on enabling you to do the following:

- Create a data LIF with the correct service policy and default route
- Ping the data LIF from the Linux client
- Use the `rpcinfo` command to get information about the RPC service

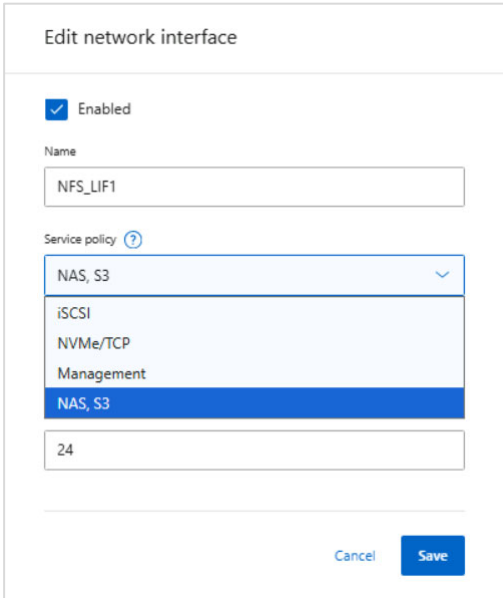
Task 1: Create Data LIF with Correct Service Policy and Default Route

In this task, you create a data LIF with the correct service policy and default route for the storage VM SVMNFS1.

Step	Action
1-1	From the NetApp ONTAP System Manager menu, select Network > Overview. 
1-2	On the Network interfaces tab, click + Add. 

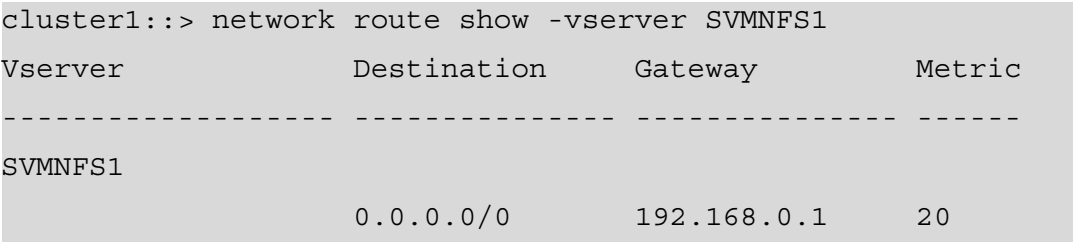
Step	Action
1-3	On the Add network interface page, enter the following details: - Interface type: Data - Protocol: NFS, SMB/CIFS, and S3 - Storage VM: SVMNFS1 - Name: NFS_LIF1 - Home node: cluster1-01 - IP address: 192.168.0.52 
1-4	Click Add optional gateway .
1-5	Click Save. 

Step	Action																				
1-6	 Optional gateway refers to configuring a default route to a gateway. Adding a gateway enables the storage VM (storage virtual machine, also known as SVM) to send traffic to external networks, making it a crucial step in network configuration when dealing with multiple subnets or external communications.																				
1-7	Observe that the new data LIF is created. Network interfacesSubnets + Add <table><tr><th>Name</th><th>Status</th><th>Storage VM</th><th>IPspace</th><th>Address</th></tr><tr><td>iscsi_lif1</td><td></td><td>san_svm</td><td>Default</td><td>192.168.0.163</td></tr><tr><td>iscsi_lif2</td><td></td><td>san_svm</td><td>Default</td><td>192.168.0.164</td></tr><tr><td>NFS_LIF1</td><td></td><td>SVMNFS1</td><td>Default</td><td>192.168.0.52</td></tr></table>	Name	Status	Storage VM	IPspace	Address	iscsi_lif1		san_svm	Default	192.168.0.163	iscsi_lif2		san_svm	Default	192.168.0.164	NFS_LIF1		SVMNFS1	Default	192.168.0.52
Name	Status	Storage VM	IPspace	Address																	
iscsi_lif1		san_svm	Default	192.168.0.163																	
iscsi_lif2		san_svm	Default	192.168.0.164																	
NFS_LIF1		SVMNFS1	Default	192.168.0.52																	
1-8	Position your cursor over NFS_LIF1, click the ellipsis (three vertical dots), and select Edit. Network interfaces + Add Name iscsi_lif1 iscsi_lif2 NFS_LIF1  Edit Migrate Delete																				

Step	Action
1-9	Observe that the Service policy is NAS, S3 and then click Cancel. 
1-10	Repeat steps 1-2 through 1-9 and create another data LIF, this one being NFS_LIF2 for the cluster1-02 node with IP address 192.168.0.53.

Task 2: Ping Data LIF from Linux Client

In this task, you verify the default route and ping the data LIF from the Linux client.

Step	Action
2-1	In the cluster management CLI, run the following command to verify the default route for the SVMNFS1 storage VM: <pre>network route show -vserver SVMNFS1</pre> 
2-2	If you see the following output, it means that a default route is not present. <pre>cluster1::> network route show -vserver SVMNFS1 There are no entries matching your query.</pre>
2-3	If your output matches the output that is shown in step 2-2, then run the following command to create a default route: <pre>network route create -vserver SVMNFS1-destination 0.0.0.0/0 -gateway 192.168.0.1</pre>

Step	Action
2-4	Omit step 2-3 if your output matches step 2-1.
2-5	From the Ubuntu client command prompt, ping the NFS_LIF1 data LIF by running the following command: <pre>ping 192.168.0.52</pre> <pre> root@ubuntu:~# ping 192.168.0.52 PING 192.168.0.52 (192.168.0.52) 56(84) bytes of data. 64 bytes from 192.168.0.52: icmp_seq=1 ttl=64 time=1.09 ms 64 bytes from 192.168.0.52: icmp_seq=2 ttl=64 time=0.388 ms 64 bytes from 192.168.0.52: icmp_seq=3 ttl=64 time=0.376 ms 64 bytes from 192.168.0.52: icmp_seq=4 ttl=64 time=0.431 ms 64 bytes from 192.168.0.52: icmp_seq=5 ttl=64 time=0.371 ms ^C --- 192.168.0.52 ping statistics --- 5 packets transmitted, 5 received, 0% packet loss, time 4081ms rtt min/avg/max/mdev = 0.371/0.531/1.092/0.280 ms </pre>
2-6	Press Ctrl+C to stop.

Task 3: Use rpcinfo Command to Get Information bout RPC Service

In this task, you verify the availability and status of the RPC service.

Step	Action
3-1	From the Ubuntu client command prompt, verify that the <code>rpcbind</code> service has started. service rpcbind status Sample output: <pre>root@ubuntu:~# service rpcbind status • rpcbind.service - RPC bind portmap service Loaded: loaded (/lib/systemd/system/rpcbind.service; enabled; vendor prese> Active: active (running) since Thu 2025-11-27 00:27:50 EST; 1 week 0 days > TriggeredBy: • rpcbind.socket Docs: man:rpcbind(8) Main PID: 722 (rpcbind) Tasks: 1 (limit: 9390) Memory: 744.0K CPU: 1.537s CGroup: /system.slice/rpcbind.service └─722 /sbin/rpcbind -f -w Nov 27 00:27:50 ubuntu systemd[1]: Starting RPC bind portmap service... Nov 27 00:27:50 ubuntu systemd[1]: Started RPC bind portmap service. lines 1-14/14 (END)</pre>
3-2	Type <code>q</code> to exit.
3-3	If <code>rpcbind</code> is not running, start it. service rpcbind start
3-4	 If <code>rpcbind</code> is already running, skip step 3-3.

End of exercise